

Foodways of the Upper Paleolithic

Food-gathering technologies of the Upper Paleolithic were substantially more varied and complex than those of early periods of human evolution. Specialized stone tools—such as fine points, sharp and durable blades, and microblades—had advantages in hunting, gathering, and processing foods. The advent of the spear thrower, or atlatl, and the bow and arrow brought even greater advantages in hunting, as they could be used from a distance, allowing dangerous and powerful large animals (including Pleistocene megafauna) to be hunted with much less risk than was involved in trying to spear the animals from close range.

Aquatic resources became another major food item. From around 150,000 years ago at the earliest, we find evidence of shellfish collected and deposited in archaeological sites in Africa; fishing became evident in the archaeological record by 100,000 years ago. Harpoons are one early fishing technology evident archaeologically, although it is the presence of fish bones in archaeological sites that is our earliest evidence of fishing. Nets appeared later in human history, around 30,000 years ago, but this may have been due to preservation biases that made these thin, organic materials much less likely to survive to the present day than hard bone and stone points.

Plant foods certainly were present in even the early human diet, but when did humans move from using raw, unprocessed resources—such as berries and edible leaves—to using underground storage organs (roots, bulbs, and tubers) and hard seeds (such as grains) that required cooking and/or other forms of processing to be made edible? These plant remains themselves often are not preserved archaeologically, so archaeologists often look to durable processing tools, such as sickles (stone tools for harvesting grasses) and grinding stones (stone slabs for grinding hard seeds into flour), as evidence of these foods. Grinding stones are present in areas where agriculture centered on cereal crops (grasses with edible seeds, such as wheat, barley, and millet) eventually arose, but from at least 10,000 years earlier. For example, the site of Ohalo II in Israel, dated to 23,000 years ago, includes abundant evidence of the intensive collection and grinding of large quantities of wild cereal grains. Similar evidence has been discovered in China. But even Neanderthals cooked and ate grass seeds, in areas as distant as northern Europe and the Zagros Mountains of Iraq, based on evidence from starch stuck in their teeth (<https://www.pnas.org/doi/10.1073/pnas.1016868108>). How widespread the use of cooked, processed plants was in both Neanderthal and modern human diets of the Upper Paleolithic is unknown, but strategies for the harvest and technologies for the processing of such foods were clearly present in both populations.

Broad-Spectrum Diets

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It is clear that humans developed more complex diets in the Late Pleistocene, drawing on an advanced Upper Paleolithic tool kit that included specialized tools for hunting, fishing, processing, and cooking animal and plant prey. In some cases, this led to hyperspecialized diets that focused on one primary prey item to the exclusion of many others. For example, in southwestern France, reindeer often dominate the animal-bone assemblages of Upper Paleolithic sites in the region. Modern human hunters focused on this prey—but why? One possible answer is that reindeer were the most common large animal on the landscape, so, frequently encountered. An alternative theory is that many prey were available, but humans preferred to hunt reindeer; presumably, the time and effort involved in the hunt were low relative to the amount of meat and other usable resources (antler, bone, bone marrow, sinew, hide) recovered from a reindeer. While we cannot always distinguish between these possibilities, it is possible to track changes over time. As climatic change led to changing landscapes that no longer favored reindeer, diets in the region changed to match; as the area became more forested toward the beginning of the Holocene, deer replaced reindeer in many archaeological assemblages.

But such replacement is not always of one preferred species for another. If preferred animals are less common or harder to catch than easier food items that yield fewer calories or resources, those less preferred animals may also be hunted when encountered. Formal models, such as the diet breadth model, explain when hunters should shift from one prey item to many. Whether this model applies widely is still debated, but evidence supports a broadening of the diet range toward the end of the Pleistocene. The question of whether this was due to expanding human populations putting more pressure on preferred animal and plant resources, or primarily due to climate change that led to differences in the availability of food items, is still under investigation. It is clear, however, that broad-spectrum diets such as these were common in many areas of the world in the latest Pleistocene, the period immediately before the advent of agriculture.